Project Design Phase
Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

Model Performance Testing – Smart Library System

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1. Data Rendered:

The library dataset, including details such as Book ID, Book Name, Author, Availability, Issue Status, and User details, was successfully loaded into the system. The data is displayed properly in tables and dashboard components without any loss or distortion.

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2. Data Preprocessing:

Data preprocessing was performed to ensure quality and consistency. Missing values were handled, duplicate book records were removed, and data types were formatted correctly (e.g., dates for issue/return and numeric values for fines). This improved the accuracy of the system.

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3. Utilization of Filters:

Filters and search options were implemented to allow users to interact with the data easily. Users can filter books based on:

- **Author**
- **Availability status**
- **Issue status**
- **Category**

This helps in quickly finding required books and improves user experience.

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4. Calculated Fields Used:

Calculated fields were created to derive meaningful insights, such as:

- **Total Books**
- **Available Books**
- **Issued Books**
- **Overdue Books**
- **Fine Amount (if applicable)**

These fields help in summarizing and analyzing library data effectively.

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5. Dashboard Design:

The dashboard is designed with a clean and user-friendly interface. It includes visual elements like bar charts, pie charts, and line graphs to represent library data clearly.

- **Number of Visualizations / Graphs: 6**

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6. Story Design:

The report/dashboard is organized to present insights step-by-step, such as:

- **Library overview**
- **Book availability analysis**
- **Issue and return trends**

This makes it easy for users to understand and interpret the data.

- **Number of Visualizations / Graphs: 5**

Conclusion:

The Smart Library system performs efficiently in terms of data handling, visualization, and user interaction. The use of preprocessing, filters, and calculated fields ensures accurate and meaningful insights, improving overall library management.